Subtraction in $\mathbb{Q}$

1. $\frac{8}{9} - \frac{2}{9}$
2. $\frac{5}{6} - \frac{1}{4}$
3. $\frac{3}{10} - \frac{4}{5}$
4. $-\frac{7}{12} - \frac{1}{3}$
5. $4\frac{1}{2} - 1\frac{5}{6}$
6. $-\frac{3}{8} - \left(-\frac{5}{12}\right)$
7. $\frac{1}{2} - \frac{2}{3} - \frac{1}{4}$
8. $-2\frac{3}{4} - \left(-5\frac{1}{3}\right)$
9. A wooden board of length $\frac{7}{8}$ meters has a piece of length $\frac{1}{3}$ meters cut off. What length of the board remains?
10. At midnight, the temperature was $-1\frac{1}{4}^{\circ}$ C. By 6:00 AM, the temperature dropped by $2\frac{5}{8}^{\circ}$ C. What was the temperature at 6:00 AM?

Solutions

1. $\frac{2}{3}$

   Explanation: Subtract numerators over common denominator 9:

   $$\frac{8}{9} - \frac{2}{9} = \frac{8-2}{9} = \frac{6}{9}.$$

   Simplify by dividing numerator and denominator by 3: $\frac{6 \div 3}{9 \div 3} = \frac{2}{3}$.

2. $\frac{7}{12}$

   Explanation: Find common denominator ($\text{LCM}(6, 4) = 12$):

   $$\frac{5}{6} = \frac{10}{12}, \quad \frac{1}{4} = \frac{3}{12}.$$

   Subtract:

   $$\frac{10}{12} - \frac{3}{12} = \frac{7}{12}.$$

3. $-\frac{1}{2}$

   Explanation: Find common denominator ($\text{LCM}(10, 5) = 10$):

   $$\frac{3}{10} - \frac{8}{10} = \frac{3-8}{10} = -\frac{5}{10} = -\frac{1}{2}.$$

4. $-\frac{11}{12}$

   Explanation: Convert $\frac{1}{3}$ to denominator 12: $\frac{1}{3} = \frac{4}{12}$.

   $$-\frac{7}{12} - \frac{4}{12} = \frac{-7-4}{12} = -\frac{11}{12}.$$

5. $\frac{8}{3}$ **or** $2\frac{2}{3}$

   Explanation: Convert mixed numbers to improper fractions:

   $$4\frac{1}{2} = \frac{9}{2}, \quad 1\frac{5}{6} = \frac{11}{6}.$$

   Common denominator 6: $\frac{9}{2} = \frac{27}{6}$.

   $$\frac{27}{6} - \frac{11}{6} = \frac{16}{6} = \frac{8}{3} = 2\frac{2}{3}.$$

6. $\frac{1}{24}$

Explanation: Subtracting a negative is equivalent to adding:

$$-\frac{3}{8} - \left(-\frac{5}{12}\right) = -\frac{3}{8} + \frac{5}{12}.$$

Common denominator ($\text{LCM}(8, 12) = 24$):

$$-\frac{9}{24} + \frac{10}{24} = \frac{-9 + 10}{24} = \frac{1}{24}.$$

7. $-\frac{5}{12}$

Explanation: Find common denominator ($\text{LCM}(2, 3, 4) = 12$):

$$\frac{6}{12} - \frac{8}{12} - \frac{3}{12} = \frac{6 - 8 - 3}{12} = -\frac{5}{12}.$$

8. $\frac{31}{12}$ **or** $2\frac{7}{12}$

Explanation: Rewrite subtract-negative as addition and convert to improper fractions:

$$-2\frac{3}{4} + 5\frac{1}{3} = -\frac{11}{4} + \frac{16}{3}.$$

Common denominator 12:

$$-\frac{33}{12} + \frac{64}{12} = \frac{-33 + 64}{12} = \frac{31}{12} = 2\frac{7}{12}.$$

9. $\frac{13}{24}$ **meters**

Explanation: Subtract cut length from total length:

$$\frac{7}{8} - \frac{1}{3} = \frac{21}{24} - \frac{8}{24} = \frac{13}{24} \text{ meters.}$$

10. $-\frac{31}{8}^{\circ}$**C or** $-3\frac{7}{8}^{\circ}$**C**

Explanation: Subtract temperature drop from initial temperature:

$$-1\frac{1}{4} - 2\frac{5}{8} = -\frac{5}{4} - \frac{21}{8}.$$

Common denominator 8:

$$-\frac{10}{8} - \frac{21}{8} = -\frac{31}{8} = -3\frac{7}{8}^{\circ}\text{C}.$$